

Theorem (Universal Approximation Theorem)

If f is a continuous function on a compact set K , then for every $\varepsilon > 0$, there exists a feed-forward network with a single hidden layer and a sufficient number of sigmoidal neurons such that the network's output is at most ε away from f .

That is:

$$\forall f \in C(K), \forall \varepsilon > 0, \exists N$$

such that

$$\sup_{x \in K} |f(x) - N(x)| < \varepsilon.$$

Appendix: Answers to the Questions

1) What is a compact set?

A set K is called compact if it satisfies two conditions:

- **Closed:** all of its limit points (boundary points) belong to the set.
- **Bounded:** it fits inside some finite interval or region.

Simple examples:

- The interval $[0, 1]$ in real numbers is compact (closed and bounded).
- The interval $(0, 1)$ is not compact (it is open).
- In 2D space, a closed disk is compact.

Why is it important?

In this theorem, compactness of K guarantees that a continuous function on it behaves “well” (for instance, it is uniformly continuous), which makes uniform approximation by a network possible.

2) What is a feed-forward network?

A feed-forward neural network is the simplest type of neural network in which information flows in only one direction:

$$\text{Input} \rightarrow \text{Hidden Layer} \rightarrow \text{Output}$$

There is no feedback from the output back to the input. If it has “one hidden layer,” it means:

$$x \rightarrow (\text{hidden layer neurons}) \rightarrow (\text{output neuron}) \rightarrow N(x)$$

Each neuron takes a linear combination of its inputs and then applies an activation function (such as a sigmoid) to it.

3) What is the function class $C(K)$?

The notation $C(K)$ refers to the set of continuous real-valued functions on the set K .

That is:

$$C(K) = \{f : K \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$$

So when we write $f \in C(K)$, it means f is a continuous function on K .

4) What is K ?

K is the domain of the function; a compact subset of Euclidean space (for example, $\mathbb{R}^n$).

Example: $K = [0, 1]^n$, the unit cube in n -dimensional space.

5) What is N ?

N is the function produced by the neural network. That is:

$$N(x) = \text{the output of the neural network for input } x$$

With specific weights and biases, the network defines a function $N : K \rightarrow \mathbb{R}$. The theorem states that this N can approximate f to within accuracy ε .

6) What is sup?

sup stands for **Supremum**, meaning the least upper bound.

For a set of numbers, sup is the smallest number that is greater than or equal to all numbers in that set.

In this theorem:

$$\sup_{x \in K} |f(x) - N(x)| < \varepsilon$$

This means: the largest distance between $f(x)$ and $N(x)$ over the entire domain K is less than ε .

This is called **uniform approximation**, because the error is controlled simultaneously at all points of K .

Summary of the Theorem in Plain Language

If f is a continuous function on a compact set such as K , then for any desired accuracy $\varepsilon > 0$, one can construct a feed-forward neural network with one hidden layer and enough sigmoidal neurons, such that the network's output $N(x)$ differs from $f(x)$ by at most ε at every point of K .

In other words: *Neural networks can approximate any continuous function to any desired accuracy.*